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RESEARCH ARTICLE

seUNet-Trans: A Simple Yet Effective UNet-Transformer Model for Medical Image Segmentation

TAN-HANH PHAM[1], XIANQI LI[2], AND KIM-DOANG NGUYEN[1]
[1]Department of Mechanical Engineering, Florida Institute of Technology, Melbourne, FL 32901, USA
[2]Department of Mathematics and Systems Engineering, Florida Institute of Technology, Melbourne, FL 32901, USA

Corresponding authors: Tan-Hanh Pham (tpham2023@my.fit.edu) and Kim-Doang Nguyen (knguyen@fit.edu)

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ABSTRACT Medical image segmentation plays a crucial role in modern clinical practice, enabling accurate diagnosis and personalized treatment plans. Advancements in machine learning, particularly deep learning techniques, have significantly driven this progress. While Convolutional Neural Networks (CNNs) dominate the field, transformer-based models are emerging as powerful alternatives for computer vision tasks. However, most existing CNN-Transformer models underutilize the full potential of Transformers, often relegating them to assistant modules. To address this issue, we propose a novel and efficient UNet-Transformer (seUNet-Trans) model for medical image segmentation. The seUNet-Trans framework leverages a UNet architecture for feature extraction, generating rich representations from input images. These features are then passed through a bridge layer that connects the UNet to a transformer module. To improve efficiency, we employ a novel pixel-wise embedding method that eliminates the need for position embedding vectors. We utilize spatially reduced attention within the transformer to reduce computational complexity. By combining the strengths of UNet's localization capabilities and the transformer's ability to capture long-range dependencies, seUNet-Trans effectively captures both local and global information within medical images. This holistic understanding enables the model to achieve superior segmentation performance. The efficacy of our model is demonstrated through extensive experimentation on seven medical image segmentation datasets. The seUNet-Trans model outperforms several state-of-the-art segmentation models, achieving impressive mean Dice Coefficient (mDC) and mean Intersection over Union (mIoU) scores. On the CVC-ClinicDB dataset, it achieves scores of 0.945 and 0.895, respectively; on the GlaS dataset, it scores 0.899 and 0.823, respectively; on the ISIC 2018 dataset, it achieves 0.922 and 0.854, respectively; and on the Data Science Bowl dataset, it scores 0.928 and 0.867, respectively. The code is available on seUnet-Trans.

INDEX TERMS Polyps, colonoscopy, medical image analysis, deep learning, vision transformers.

I. INTRODUCTION

Medical image segmentation involves identifying and extracting meaningful information from complex medical images, playing a crucial step in many clinical applications including computer-aided diagnosis, image-guided surgery, and treatment planning [1], [2]. To date, manual segmentation by trained experts such as radiologists or pathologists remains the gold standard for delineating anatomical structures and pathological abnormalities. However, this process is costly, labor-intensive, and often requires significant experience. For example, differentiating between benign and malignant skin lesions in dermoscopic images is essential for early and accurate skin cancer diagnosis. Another critical application is differentiating subtle variations in brain tissue from MRI scans, which is crucial for diagnosing neurological diseases such as Alzheimer's or multiple sclerosis. Traditionally, these tasks rely on manual segmentation by neurologists and

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visual examination by dermatologists. Manual segmentation is a time-consuming and subjective process, and visual examination can be prone to human error.

Automated segmentation techniques, particularly those based on deep learning algorithms, have emerged as valuable tools in medical imaging. Deep learning models can be trained on vast datasets of labeled dermoscopic images, enabling them to learn the subtle color, texture, and shape variations that distinguish cancerous from benign lesions. This not only assists dermatologists in early detection but also improves the accuracy and consistency of skin cancer diagnosis. Consequently, there is a significant demand for deep learning-driven automated medical image segmentation in clinical practice.

As a prominent subset of various image segmentation models, convolutional neural networks (CNN) have proven to be highly effective and promising in numerous medical image segmentation tasks. One notable example is UNet, a type of fully convolutional network (FCN) [3], [4], [5], [6], [7]. UNet consists of a symmetric encoder-decoder architecture with skip connections that pass features from the encoder path to the decoder path. However, due to its limited ability to capture the long-range dependencies and global context information in images, this architecture often produces inferior performance. This limitation is particularly evident when dealing with target information that exhibits significant differences among patients in texture, shape, and size.

To address these shortcomings, recent research suggests implementing self-attention mechanisms [8], [9] to allow the network to focus on relevant image regions, improving its ability to capture long-range dependencies between structures or subtle texture variations crucial for segmentation. Examples include convolutional self-attention modules, as proposed in [10], [11] or transformer-based architectures like Medical Transformer [12]. Additionally, self-attention mechanisms can improve robustness to noise and handle complex anatomical shapes [13]. Other approaches are also being explored to enhance segmentation performance, for instance, incorporating prior anatomical knowledge as a guiding force through techniques like anatomical guidance modules [14] or integrating with other modalities, such as clinical data. These methods provide additional context and guidance, thereby improving the accuracy and reliability of segmentation models.

The transformer concept has emerged as an alternative architecture that entirely abandons convolutional operators and relies exclusively on attention mechanisms [15]. Initially conceived for sequence-to-sequence tasks in natural language processing (NLP), Transformers have recently sparked significant debate within the computer vision (CV) community. Unlike previous CNN-driven methods, transformers excel at capturing global context information and demonstrate enhanced adaptability for downstream tasks when pre-trained on a large scale. For instance, the first fully self-attention-based vision transformer (ViT) for image recognition was introduced in [16], achieving competitive outcomes on ImageNet [17] using 2D image patches with positional embedding as an input sequence, provided that it was pre-trained on an extensive external dataset.

Work in [18] employs a transformer-based approach as a fully end-to-end object detector (DERT), delving into the connections between objects and the overall image context for object detection. Segmentation Transformer (SETR) replaces the traditional encoders with transformers in the standard encoder-decoder networks, effectively attaining state-of-the-art (SOTA) outcomes in the task of natural image segmentation [19]. There are several transformer-based models designed for a specific task [20], [21], [22], [23]. While the transformer method is good at capturing global context, it struggles to grasp fine-grained details, especially for medical images. To overcome this limitation, efforts have been made by researchers to integrate CNN- and Transformer-based models into each other [24], [25], [26], [27], [28]. In particular, TransUNet [29] and TransFuse [30] are representative examples of combining the transformer and UNet for medical image segmentation.

A. MOTIVATION

While progress has been made, current methods for medical image segmentation underutilize the full potential of transformers. These approaches typically rely heavily on Unet CNNs for feature extraction, relegating the transformer to an assistant role. This limits the model's ability to fully exploit the transformer's strength in capturing long-range dependencies and global context. To address this limitation, we propose a novel and integrative UNet-Transformer model, named seUNet-Trans.

B. NOVELTY

The novelty of seUNet-Trans lies in its architecture and its capability of using transformer networks to their fullest potential. In the proposed model, the UNet functions as a feature extractor, generating rich representations from the input images. These features are then passed through a strategically designed bridge layer that connects the UNet module to the transformer module. We employ a pixel-level embedding technique within the transformer to enhance efficiency, eliminating the need for position embedding vectors. Unlike traditional approaches, in our framework the transformer plays a central role in modeling relationships and dependencies among the input features, ultimately generating a segmentation map for the input image. By leveraging the strengths of UNet and the transformer, seUNet-Trans effectively captures both local and global information within the image, while also modeling long-range dependencies between image elements.

The rest of the paper is organized as follows. Section II provides an overview of related work in the domain of medical image segmentation. Section III presents the architecture

of the proposed seUNet-Trans model. The training data and evaluation metrics are described in Section IV. Section V focuses on numerical experiments and comparisons with other state-of-the-art segmentation models. Section VI concludes our work.

II. RELATED WORK

In this section, we begin by providing an overview of the commonly used CNN-based methods for medical image segmentation. Recent advancements in the application of transformers will be explored within the realm of computer vision, particularly in segmentation tasks. Finally, we highlight the standard techniques that merge both CNN and Transformer architectures.

A. CNN-BASED MEDICAL IMAGE SEGMENTATION

The field of medical image segmentation has witnessed remarkable achievements using CNNs, especially the FCN, UNet, and their variants. These variants constantly push the boundaries by introducing innovative techniques. For instance, UNet++ introduces a set of nested and densely skip connections to minimize the discrepancy between the encoding and decoding process [31]. Attention U-Net proposes an innovative attention gate method, which empowers the model to prioritize targets with varying sizes and exclude non-pertinent feature responses [32]. Res-UNet incorporates a weighted attention mechanism and a skip connection scheme to enhance the performance of retinal vessel segmentation [33], [34]. R2U-Net merges the advantages of residual networks with UNet to elevate its feature representation capabilities [35].

Beyond these specific examples, architectures like PraNet with its parallel partial decoder and reverse attention model have been successful in polyp segmentation [36]. KiU-Net demonstrates improved segmentation of small anatomical structures by utilizing both under-complete and over-complete features [37]. DoubleU-Net establishes a robust foundation for medical image segmentation by chaining two U-Nets and implementing atrous spatial pyramid pooling (ASPP) [38]. FANet further enhances performance by incorporating the mask from the previous epoch into the current feature map [39]. Given that these methods are anchored in CNNs, they inherently miss out on capturing long-range dependencies and understanding global contextual ties.

B. TRANSFORMER-BASED MEDICAL IMAGE SEGMENTATION

Transformers were first developed for machine translations and have now achieved top-tier performance in various NLP tasks [15]. Inspired by their successes, many efforts have been made to adapt Transformers for computer vision tasks. In particular, ViT is the pioneering endeavor demonstrating that a solely transformer-based architecture can attain superior performance in image recognition, given pre-training on a substantial dataset [16].

Utilizing ViT as an encoder, Segmenter provides a segmentation framework by proposing a mask transformer decoder to generate class embeddings [40]. With a combination of a transformer-based hierarchical encoder and a lightweight multilayer perceptron (MLP), SegFormer offers a simple yet potent segmentation architecture [41]. By integrating an additional control function into the self-attention module, MedT proposed a gated axial-attention that extends the existing transformer-based architecture [12].

Swin Transformer recently attracted great attention due to its exceptional performance on a number of benchmarks for tasks such as image classification, object detection, and semantic segmentation [42]. In contrast to many previous transformer-based models, Swin Transformer proposes a hierarchical architecture whose representation is computed with shifted windows. This strategy enhances efficiency by restricting self-attention computation to non-overlapping local windows while also allowing for cross-window connection. The hierarchical structure combined with the shifted window technique as a backbone can benefit other network architecture. By incorporating Swin Transformer into the encoder and decoder of the U-shaped architecture, DS-TransUNet proposes a novel deep medical image segmentation framework that can effectively capture the non-local dependencies and multiscale contexts for improving the semantic segmentation quality of varying medical images [43]. Extensive numerical experiments across seven typical medical image segmentation tasks show the effectiveness of this framework.

Despite transformer-based methods that can model the global context at all stages, they process inputs as 1D sequences which may result in low-resolution features, thereby lacking precise localization information. Simply resorting to direct upsampling to achieve full resolution doesn't effectively recover this information which therefore leads to an imprecise segmentation result. To address this issue, significant research efforts have been made to integrate CNN with the self-attention mechanism by characterizing the global relationships of all pixels through the feature maps. TransUNet is the first such framework combining Transformer with UNet and achieving SOTA performance on medical image segmentation tasks [29]. TransFuse proposes a shallow CNN-based encoder and transformer-based segmentation network in parallel to enhance the efficiency for modeling global contexts [30]. For 3D segmentation, UNETR is formulated by using a Transformer in the encoder to learn sequence representations of the input volume [44]. To reduce the complexity of the attention mechanism in the Transformer, UNETR++ was designed in [45] by leveraging paired attention blocks.

Furthermore, work in [46] proposed nnFormer to improve medical segmentation performance by replacing the concatenating connection in the UNet model by using skip attention. A CNN-Transformer architecture was developed in [47] by designing two encoders using Swin Transformers

and Residual CNNs, then blending them, and generating output. Compared to the existing work on medical image segmentation, our SeUNet-Trans inherits the advantages of UNet and Transformer through its designed architecture. The Transformer head processes sequence-to-sequence fine features from the UNet encoder to generate segmentation maps.

III. METHODOLOGY

In this section, we introduce our proposed model for medical image segmentation in detail. Our model is composed of a UNet structure as a backbone coupled with a Transformer head. The U-shaped backbone designed with an encoder and a decoder processes input images to produce multiple feature maps, which are then fed into specially designed bridge layers. Subsequently, the Transformer head processes the output from these bridge layers to yield the final prediction. The architecture of the seUNet-Trans model is depicted in Fig. 1. The rest of this section elaborates on the working principles of the model's components, following a top-down approach.

A. ENCODER

The encoder's role is to extract features from the input images within the network. This is achieved through a series of convolutional layers, referred to as UNet blocks, succeeded by max-pooling layers. As the input images progress through these UNet blocks, their spatial dimensions are reduced, while the depth (or the number of channels) of the feature maps increases. Based on the foundational work that proposes UNet, we constructed the encoder module with four UNet blocks. The specific design of the UNet block is illustrated in Fig. 2.

The UNet block includes two convolutional neural networks (Conv) [48] followed by a batch normalization function [49] and a rectified linear unit ReLU activation function [50]. The structure of the UNet block can be formulated as:

$$\begin{aligned} \hat{F}_i &= \mathrm{ReLU}\left(\mathrm{Batch}\left(\mathrm{Conv}_{(C_{in},C_h)}(F_{i-1})\right)\right), \\ F_i &= \mathrm{ReLU}\left(\mathrm{Batch}\left(\mathrm{Conv}_{(C_h,C_o)}(\hat{F}_i)\right)\right), \quad \forall i \geq 1, \end{aligned} \tag{1}$$

where $\hat{F}_i$ and F_i are intermediate and final feature maps for each UNet block. The terms C_{in}, C_h, C_o represent input, hidden, and output layers, respectively.

B. DECODER

The decoder component focuses on upsampling the encoded feature maps back to the input image size and mirrors the architecture of the encoder. Instead of max-pooling layers, it uses an up-convolution (or transpose convolution) to increase the spatial dimensions. Skip connections, a critical component of UNet, are also implemented and integrated to help the decoder retrieve spatial information lost during encoding. At each level in the decoder, the output from the corresponding encoder level (before pooling) is concatenated with the upsampled feature maps. After the skip connection, the concatenated features are passed through convolutional layers to refine the upscaled features.

Distinct from the conventional decoder's final layer, which features a single channel for binary segmentation or several channels for multi-class segmentation with each channel representing the probability of a pixel belonging to a certain class, our enhanced layer outputs multiple feature maps, which will be fed into the next specially designed component, called "bridge layers".

C. BRIDGE LAYERS

At the end of the decoder, the input images undergo a transforming process, yielding high-level features with identical dimensions but a different number of channels. To refine and expand these features, a convolution layer is employed with a kernel size of 1 and a batch size of 1. These layers act as the bridge between two intrinsically different machine learning mechanisms, i.e. the UNet and the Transformer head.

D. TRANSFORMER HEAD

The Transformer head begins by merging the features from the bridge layers. Subsequently, these merged features are flattened into sequences, and fed into the multi-head attention (MHA) mechanism. After that, the output of the MHA is passed through the multi-layer perceptron (MLP), used for mapping the input features to output features. Eventually, the output from the MLP is linearly upsampled, and processed by convolutional layers followed by BatchNorm and ReLu activation functions (CBR block) before outputting the final prediction. The structure of the Transformer head is shown in Fig. 3.

1) FEATURE EMBEDDING

The bridge layers with the size of (H, W, C_b), height, width, and number of the bridge channels, are merged by using a convolutional layer with the kernel size E, stride S, and padding P are 3, 4, and 1 respectively. After passing through the convolutional layer, the output resolution of the bridge layers is computed as:

$$\begin{aligned} H_{out} &= \frac{(H-E+2P)}{S}+1, \\ W_{out} &= \frac{(W-E+2P)}{S}+1. \end{aligned} \tag{2}$$

In the context of image segmentation, our objective is to establish the relationship between pixels in the image. This can be accomplished through various methods, such as CNN-based techniques, attention mechanisms, and graph neural networks [16], [48], [51]. In this particular study, we utilize the attention mechanism due to its effectiveness in capturing long-range features. We treat each pixel and its variations across different spatial dimensions (represented by various features in different channels) as a single input vector denoted as a. In other words, the merging features are flattened into

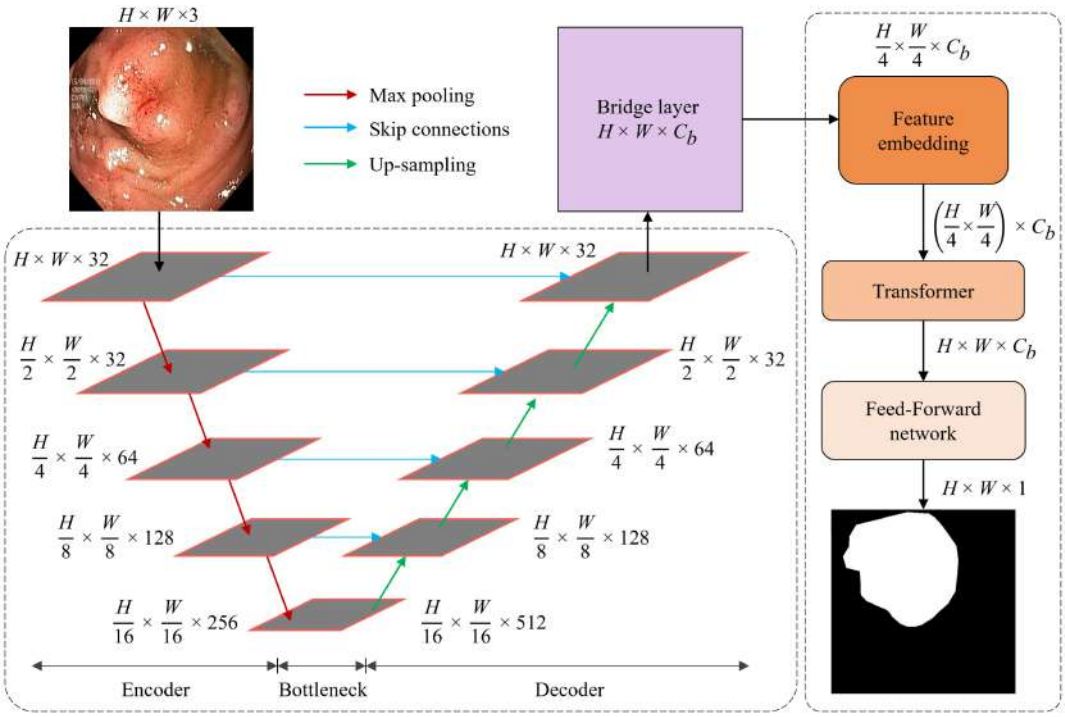


FIGURE 1. The architecture of the seUNet-Trans consists of a Unet model on the left-hand side and a Transformer model on the right-hand side.

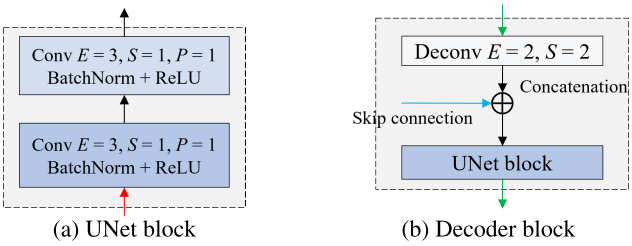


FIGURE 2. UNet and Decoder blocks in the seUnet-Trans model.

sequences, and the dimensions of the sequences are $A \in \mathbb{R}^{N \times C_b}$, where $N = H_{out} \times W_{out}$.

Different from the vision transformer [16], in this study, we opted not to use position embedding vectors during the input image flattening process. This choice is grounded in our approach of merging the input image and embedding the resultant features at the pixel level. Typically, the process of merging and embedding the bridge features into sequences can be formulated as:

$$F_f = \text{Flatten}\left(\text{Conv}_{(C_b, C_b)}\left(F_b\right)\right). \tag{3}$$

Here, F_b represents bridge layers, and F_f is the embedding features.

2) TRANSFORMER BLOCK

The Transformer block consists of multi-head attention (MHA), multi-layer perceptron, Layer Norm (LN), and residual connections, and it can be formulated as

$$\begin{aligned} \hat{F}_i &= \text{MHA}\left(\text{LN}\left(F_{i-1}\right)\right) + F_{i-1}, \\ F_i &= \text{MLP}(\text{LN}(\hat{F}_i)) + \hat{F}_i. \end{aligned} \tag{4}$$

Again, $\hat{F}_i$ and F_i are intermediate and output layers of the ith Transformer block. For the first Transformer block, i.e. $i = 1$, the input is the embedding features (F_f).

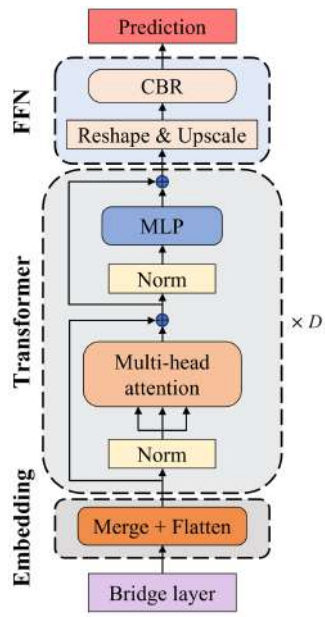


FIGURE 3. Attention head in the seUNet-Trans model.

In the MHA, the dependencies between sequences are computed by using cross-attention. In this step, the computational complexity is N^2 with N as the number of input sequences. To reduce the computation, we used the sequence reduction technique implemented in [41] and [52], making it adaptable for high-resolution input images. Therefore, the complexity becomes N^2/R, where R is the reduction rate.

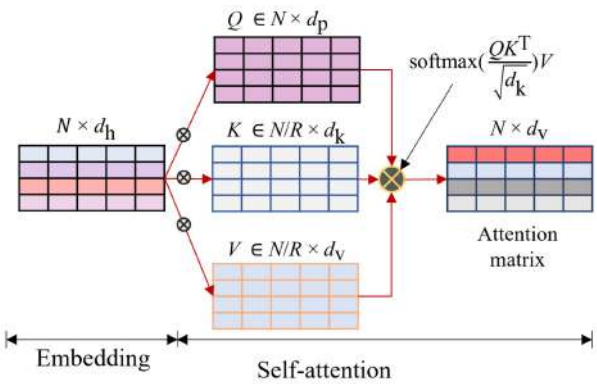


FIGURE 4. Attention mechanism.

The input sequences are divided into multiple heads h in the MHA, in which the dimension of each head is d_h, $d_h = d_N/h$. In this study, we employ the length of the embedding vector $d_N = 64$, and the number of heads is $h = 4$. The attention in each head is calculated as

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_h}})V, \tag{5}$$

in which $Q \in \mathbb{R}^{N \times d_q}$, $K \in \mathbb{R}^{N/R \times d_k}$, $V \in \mathbb{R}^{N/R \times d_v}$, and $d_q = d_k = d_v = d_h$.

Once the attention of each head is calculated, we combine all of them together to obtain the final attention matrix,

$$\begin{aligned} \text{MultiHead}(Q, K, V) &= \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W^O, \\ \text{and head}_e &= \text{Attention}(QW_e^Q, KW_e^K, VW_e^V), \end{aligned} \tag{6}$$

where, $W_e^Q \in \mathbb{R}^{d_N \times d_h}$, $W_e^K \in \mathbb{R}^{d_N/R \times d_h}$, and $W_e^V \in \mathbb{R}^{d_N/R \times d_h}$ are parameter matrices over a head, and $W^O \in \mathbb{R}^{h.d_v \times d_N}$ is the total parameter matrix. Fig. 4 illustrates the attention mechanism intuitively.

The output from the MHA is added to its input through a residual connection. This connection facilitates the network's ability to learn residual information, representing the discrepancy between the expected output and the current estimate. Consequently, the network can adeptly capture and distribute gradient information during training, even in profoundly deep networks. Such a mechanism aids in efficiently training deeper neural architectures while addressing the vanishing gradient challenge.

Beyond the MHA, a Transformer block also encompasses a connected feed-forward network or MLP, which consists of two linear transformations with a GeLU activation [53] between them. The aggregated features are first normalized before feeding into the MLP. Similar to the output from MHA, here we used another residual connection to add the MLP's output to its input.

Equation 4 describes the MHA and MLP procedure, in which the input features are mapped to the output features following the standard transformer [15]. The process of the transformer block can be repeated D times, and in this study, we choose $D = 3$.

3) FEED-FORWARD NETWORK

The Feed-Forward Network (FFN) takes in the embedded sequences from the Transformer block to extract features and generate a prediction map. Given that the FFN operates on sequences as inputs, it becomes necessary to reshape these inputs to conform to the desired input shape (H_{out}, W_{out}, C_b).

Furthermore, as computed in Section III-D1, the input shape undergoes a merging operation, resulting in a fourfold reduction in size. Consequently, the reshaped features must be upsampled by a factor of four to match the original input shape (H, W, C_b). This upsampling process employs a bilinear interpolation function to increase the resolution of the feature maps. Mathematically, this step can be represented as follows:

$$F_{rs} = \text{Upscale}\left(\text{Reshape}\left(F_D\right)\right). \tag{7}$$

Here, F_{rs} represents the upsampled feature maps after reshaping, and F_D is the features from the Transformer block D, the final Transformer block.

After getting the upsampled features, they are fed into the CBR block for further processing, ultimately yielding the final prediction map. The CBR block, named for its convolutional layers, batch normalization, and ReLU activation, plays a vital role in feature refinement and spatial enhancement, enabling the network to capture intricate patterns and relationships within the data. The CBR consists of three convolutional layers, in which the first two layers with kernels size E of 3×3 are followed by batch normalization and ReLU activation, while the third layer with a kernel size E of 1×1 takes in features from previous layers and directly outputs the final prediction map M. Mathematically, this can be represented as follows:

$$\begin{aligned} \hat{F} &= \text{ReLU}\left(\text{Batch}\left(\text{Conv}_{(C_b, C_{h1})}(F_{rs})\right)\right), \\ \hat{F} &= \text{ReLU}\left(\text{Batch}\left(\text{Conv}_{(C_{h1}, C_{h2})}(\hat{F})\right)\right), \\ M &= \text{Conv}_{(C_{h2}, 1)}(\hat{F}). \end{aligned} \tag{8}$$

Again, $\hat{F}$ is the intermediate output of the CBR block. C_{h1} and C_{h2} are the hidden 1st and 2nd convolutional layers, respectively. In this study, we build the seUNet-Trans models for medical image segmentation. Hence, the final prediction M is the binary image (one class).

IV. TRAINING DATA AND EVALUATION

This section presents a comparative analysis of our proposed model with respect to SOTA models in medical image

segmentation using publicly available datasets. We first describe the datasets and outline the employed metrics to evaluate the model's efficacy. Further, specifics regarding the training and optimization processes are detailed in this section's conclusion.

A. DATASET

The seUNet-Trans models were trained on the Polyp Segmentation (Kvasir-SEG, CVC-ClinicDB, CVC-ColonDB, EndoScene), ISIC 2018, GlaS, and 2018 Data Science Bowl datasets, which are widely recognized and frequently used for evaluating various medical segmentation models such as DS-TransUNet, PraNet, and ColonSegNet [36], [43], [54].

TABLE 1. Published datasets for training and testing the seUNet-Trans models.

Dataset		Size	Training images	Test images
Kvasir-SEG		512×512	880	120
CVC-ClinicDB		384×384	550	62
Mixing Polyp segmentation	Kvasir-SEG	384×384	900	100
	CVC-ClinicDB		550	62
	CVC-ColonDB		0	380
	EndoScene		0	60
GlaS		128×128	85	80
ISIC 2018		256×256	2075	519
2018 Data Science Bowl		256×256	536	134

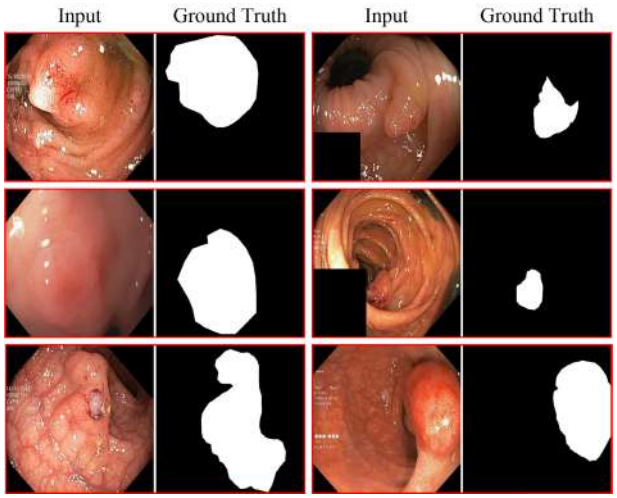


FIGURE 5. Visual illustration of input images and their corresponding ground truths (segmentation maps) from the Kvasir-SEG dataset.

For data preprocessing, we first standardized the dataset by resizing the images to a uniform scale. We then divided the pre-processed dataset into separate training and test datasets. Table 1 provides a comprehensive overview of the image divisions and the resized resolutions for various training scenarios. Notably, when dealing with the mixing Polyp segmentation case, we combined different datasets for training and testing tasks. In particular, the training set comprises 900 Kvasir-SEG images and 550 CVC-ClinicDB images, while the test set includes 100 Kvasir-SEG images, 62 CVC-ClinicDB images, 380 CVC-ColonDB images, and 60 EndoScene images following [43].

In addition to the mixing Polyp segmentation, we have conducted training for our proposed seUNet-Trans models using the Kvasir-SEG or CVC-ClinicDB datasets separately. The Kvasir-SEG dataset is partitioned into 880 images for training and 120 for testing, and the CVC-ClinicDB dataset contains 550 images for training with a test set of 62 images. Visual representations from the Kvasir-SEG dataset are displayed in Fig. 5, where the input is presented as an RGB image, and the corresponding output is a binary segmentation mask. For the GlaS, ISIC 2018, and 2018 Data Science Bowl datasets, the training dataset contains 85, 2075, and 536 images, respectively, while the test dataset comprises 80, 519, and 134 images for each.

B. MODEL TRAINING

Our seUNet-Trans models have been developed utilizing the PyTorch 1.13.1 deep learning framework. For the training process, we employ the AI.Panther high-performance computing infrastructure, equipped with A100 SXM4 GPUs, hosted at the Florida Institute of Technology.

As mentioned in section III-D3, the final prediction is the binary image, therefore we use the binary cross-entropy (BCE) loss as the objective function during the training process. In addition, we employ the Adam optimizer [55] with a learning rate of 0.0001, weight decay of 0.0001, and a batch size of 4 for model training. The model is trained up to 500 epochs, and checkpoints are saved every 10 epochs and are subsequently loaded to evaluate the model on the test dataset upon completion of the training phase.

V. EXPERIMENTAL RESULTS

This section details the performance of our model across all datasets mentioned in the last section, providing a thorough comparative analysis with SOTA models. We present our findings in tabular form and include SOTA results referenced from [43] to facilitate a clear and comprehensive comparison. For a visual illustration, Figures 6-9 display predicted results by seUNet-Trans models for a selection of representative images, along with those produced by SOTA models. To quantitatively evaluate our model's performance, tables 2-6 present the calculated metric scores obtained across the various datasets.

As mentioned in Section III-D2, we implemented a sequence reduction technique to mitigate the computational complexity in the attention head. Specifically, we adopted the reduction ratio R with values of 4, 2, and 1, which correspond to the stride S of 2, 4, and 8, respectively. It's worth noting that a smaller stride retains more information but increases the number of input sequences, which can lead to longer training time. With the pairs of reduction ratio and stride, we developed three distinct models: seUNet-Trans-L,

seUNet-Trans-M, and seUNet-Trans-S. Therefore, in this section, we compare all our models' performance with the SOTA models.

A. RESULTS ON THE KVASIR-SEG DATASET

Figure 6a presents the results predicted by our seUNet-Trans-M on the Kvasir-SEG dataset as compared to the SOTA results. We then perform a comparative analysis of these predictions against those produced by other models. The predictions of our models are comparable to those of other models, closely aligning with the objects in the ground truth. Furthermore, the calculated metric values are summarized in Table 2, revealing that the seUNet-Trans models achieve impressive values of 0.919 for mDC, 0.850 for mIoU, 0.912 for mRec., and 0.926 for mPre. Notably, seUNet-Trans outperforms other models in terms of mDC, and mPre., demonstrating superior performance. However, it's worth mentioning that the mIoU and mRec. scores of our models are relatively smaller than those of PraNet, HarDNet-MSEG, and DS-TransUNet. In addition, seUNet-Trans-M demonstrates consistent performance among our models.

TABLE 2. Quantitative results of evaluation metrics for seUNet-Trans in comparison to SOTA models on the Kvasir-SEG.

Methodology	mDC	mIoU	mRec.	mPre.
U-Net [3]	0.597	0.471	0.617	0.672
Res-UNet [56]	0.690	0.572	0.725	0.745
ResUNet++ [57]	0.714	0.613	0.742	0.784
DoubleU-Net [38]	0.813	0.733	0.840	0.861
FCN8 [4]	0.831	0.737	0.835	0.882
PSPNet [58]	0.841	0.744	0.836	0.890
HRNet [59]	0.845	0.759	0.859	0.878
DeepLabv3+ [60]	0.864	0.786	0.859	0.906
FANet [39]	0.880	0.810	0.906	0.901
HarDNet-MSEG [61]	0.904	0.848	0.923	0.907
DS-TransUNet-L [43]	0.913	**0.859**	**0.936**	0.916
seUNet-Trans-L (ours)	0.914	0.841	0.901	**0.927**
seUNet-Trans-M (ours)	**0.919**	0.850	0.912	0.926
seUNet-Trans-S (ours)	0.904	0.825	0.896	0.912

B. RESULTS ON THE CVC-CLINICDB DATASET

Figure 6b shows the predicted results by our seUNet-Trans-M versus other models. As observed, the predictions by seUNet-Trans surpass not only those of the standard UNet model but also outperform the SOTA models. A detailed comparison of the results is tabulated in Table 3, where the seUNet-Trans-M achieves remarkable performance, including a mDC of 0.945, mIoU of 0.895, mPre. of 0.951, and mRec. of 0.950. In contrast, the standard UNet model yields lower metric values with mDC, mIoU, mPre., and mRec. values of 0.872, 0.804, 0.868, and 0.917, respectively. This comparison underscores the superior performance of the seUNet-Trans on the CVC-ClinicDB dataset when compared to the baseline UNet model and other SOTA models.

C. RESULTS ON THE GLAS DATASET

Figure 7 displays the predictions by our seUNet-Trans-M on the GlaS dataset. Compared to other SOTA models, seUNet-Trans-M stands out for its robust performance in gland segmentation. In addition, the results show that it not only outperforms most of the other models but also demonstrates comparability with DS-TranUNet. In terms of visualization, our model's predictions exhibit significantly higher accuracy (indicated by the red boxes) and demonstrate fewer outliers (highlighted by the cyan boxes).

TABLE 3. Quantitative results of evaluation metrics for seUNet-Trans in comparison to SOTA models on the CVC-ClinicDB.

Methodology	mDC	mIoU	mRec.	mPre.
SFA [62]	0.700	0.607	-	-
ResUNet-mod [63]	0.779	0.455	0.668	0.888
UNet++ [31]	0.794	0.729	-	-
U-Net [3]	0.872	0.804	0.868	0.917
PraNet [36]	0.899	0.849	-	-
DoubleU-Net [38]	0.924	0.861	0.846	**0.959**
FANet [39]	0.936	0.894	0.934	0.940
DS-TransUNet-L [43]	0.942	0.894	0.950	0.937
DCSAU-Net [64]	0.916	0.861	0.920	**0.917**
seUNet-Trans-L (ours)	0.936	0.879	0.941	0.933
seUNet-Trans-M (ours)	**0.945**	**0.895**	**0.951**	0.950
seUNet-Trans-S (ours)	0.938	0.888	0.936	0.945

TABLE 4. Quantitative results of evaluation metrics for seUNet-Trans in comparison to SOTA models on the GlaS.

Methodology	mDC	mIoU	mRec.	mPre.
Seg-Net [65]	0.786	0.660	-	-
U-Net [3]	0.796	0.672	0.845	0.778
MedT [12]	0.81	0.696	-	-
UNet++ [31]	0.813	0.696	0.857	0.798
Attention UNet [32]	0.816	0.701	0.844	0.813
KiU-Net [37]	0.833	0.728	**0.889**	0.809
DS-TransUNet-L [43]	0.878	0.791	0.888	0.878
seUNet-Trans-L (ours)	0.890	0.810	0.868	**0.923**
seUNet-Trans-M (ours)	**0.899**	**0.823**	0.886	0.920
seUNet-Trans-S (ours)	0.881	0.795	0.873	0.900

As mentioned in section IV-A, although the number of training samples in the GlaS dataset is very limited, our models still perform well compared to other models. Table 4 further reinforces seUNet-Trans's proficiency, showing that its metric values surpass those of the SOTA models. Specifically, the seUNet-Trans-M achieves mDC and mIoU scores of 0.899 and 0.823, respectively, indicating its efficacy in gland segmentation on the GlaS dataset. Additionally, all three variants of our models consistently yield superior results compared to the other models. While the seUNet-Trans model does not produce outliers like the KiU-Net, it struggles to predict ambiguous gland areas compared to the DS-TransUNet.

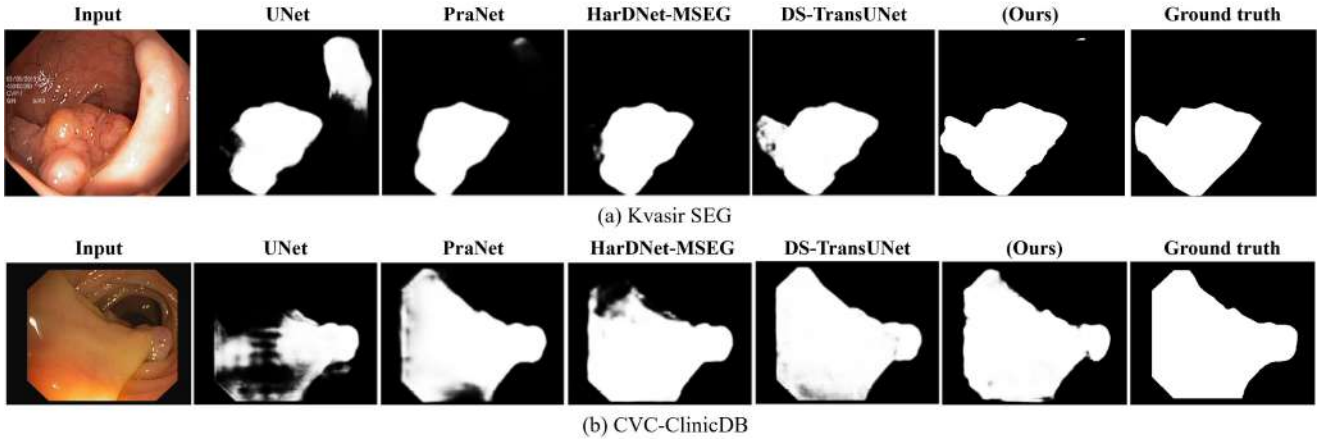


FIGURE 6. Visualization of predictions of seUNet-Trans-M and the SOTA on the Kvasir-SEG and ClinicDB datasets. These images are partially taken from [43] for comparison purposes.

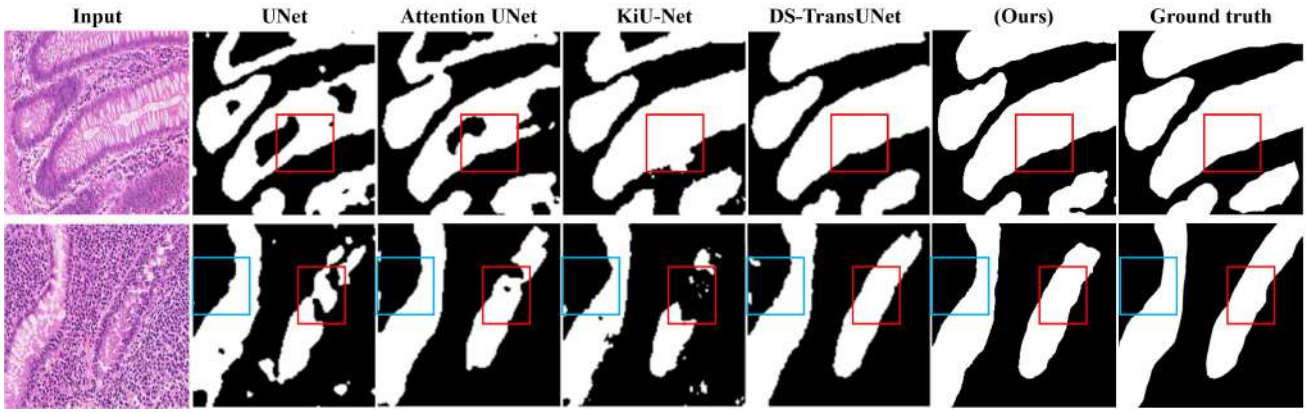


FIGURE 7. Visualization of predictions of seUNet-Trans-M and the SOTA on the GlaS dataset. In these images, accurate regions are highlighted with red rectangles, while outliers among the models are indicated with cyan rectangles. These images are partially taken from [43] for comparison purposes.

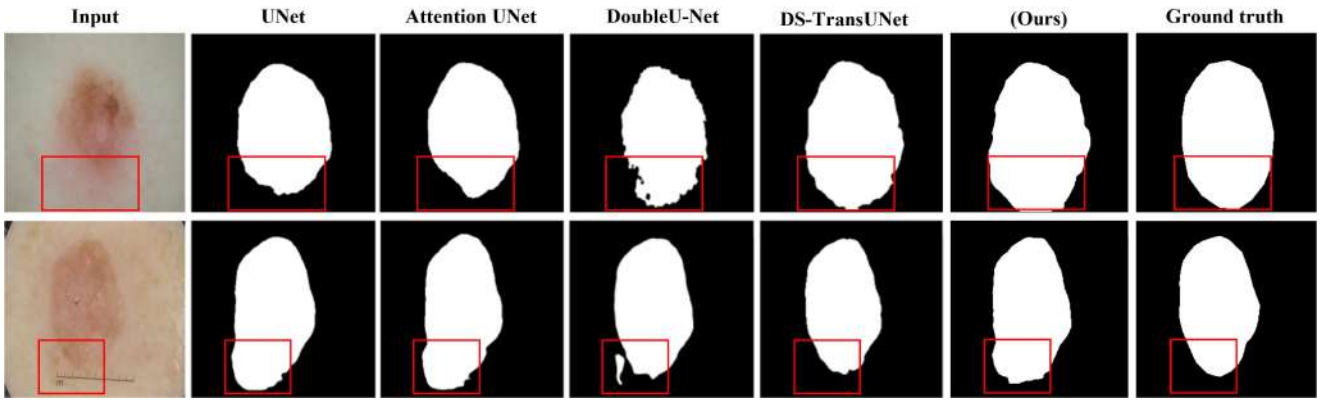


FIGURE 8. Visualization of predictions of seUNet-Trans-M and the SOTA on the ISIC2018 dataset. These images are partially taken from [43] for comparison purposes. In these images, comparison regions are highlighted with red rectangles.

D. RESULTS ON THE ISIC 2018 DATASET

Figure 8 presents the predictions by our model on the ISIC 2018 dataset, and the corresponding metrics are detailed in Table 5. In comparison to SOTA models, our models demonstrate strong performance on representative images. Although our model's predictions do not perfectly match the segmentation mask (ground truth), they represent the segmentation mask more accurately than the UNet, Attention

UNet, and DoubleU-Net. For example, in the second sample (second row in Fig. 8), our prediction covers the area below the ruler (highlighted in the red rectangles) more accurately, avoiding the extensive over-segmentation seen in UNet and Attention UNet. Additionally, it does not contain the outliers present in the DoubleU-Net predictions.

TABLE 5. Quantitative results of evaluation metrics for seUNet-Trans in comparison to SOTA models on the ISIC2018.

Methodology	mDC	mIoU	mRec.	mPre.
U-Net [3]	0.674	0.549	0.708	-
Attention U-Net [32]	0.665	0.566	0.717	-
R2U-Net [35]	0.679	0.581	0.792	-
Attention R2U-Net [35]	0.691	0.592	0.726	-
BCDU-Net (d=3) [66]	0.851	-	0.785	-
FANet [39]	0.8731	0.802	0.865	0.924
DoubleU-Net [38]	0.896	0.821	0.878	**0.946**
DS-TransUNet-L [43]	0.913	0.852	**0.922**	0.927
DCSAU-Net [64]	0.904	0.841	0.922	**0.922**
seUNet-Trans-L (ours)	0.918	0.849	0.900	0.938
seUNet-Trans-M (ours)	**0.922**	**0.854**	0.903	0.941
seUNet-Trans-S (ours)	0.921	**0.854**	0.906	0.937

In terms of the metric values, the seUNet-Trans-M still achieves commendable scores on ISIC 2018, with mDC, mIoU, mRec., and mPre. standing at 0.922, 0.854, 0.903, and 0.941, respectively. These metrics demonstrate the model's strong performance as compared to the SOTA models.

E. RESULTS ON THE 2018 DATA SCIENCE BOWL DATASET

The results of our seUNet-Trans-M on the 2018 Data Science Bowl dataset are visualized in Fig. 9. Similar to Fig.7, we compared the predicted masks generated by our models with those of the SOTA models. In this context, the red rectangles draw attention to precision, while the cyan rectangles highlight outliers. As illustrated, our model's predictions do not suffer from outliers and are more accurate compared to others.

Table 6 summarizes quantitative metrics among models. Once again, our seUNet-Trans-M demonstrates its stability across the three variants and its superiority over SOTA models. Typically, the mDC and mIoU scores of seUNet-Trans-M are 0.928 and 0.867, relatively higher than the SOTA models. In comparison, DS-TransUNet exhibits slightly lower mDC and mIoU scores of 0.922 and 0.861, respectively. This underscores the efficacy of seUNet-Trans on the 2018 Data Science Bowl dataset, as its predictions are notably free of outliers. However, our model still struggles to provide separated regions for nuclei that are close to each other, similar to other models.

F. RESULTS ON MIXED POLYP SEGMENTATION

When training and testing with individual datasets, it becomes evident that seUNet-Trans-M consistently yields reliable results. Therefore, for this experiment, we have employed this model to train and evaluate a *mixed* dataset.

As described in section IV-A, our seUNet-Trans-M was trained on a combined dataset composed of four distinct datasets for the mixed Polyp segmentation case. Figure 10 illustrates our model's performance on the test set, in which its mDC and mIoU surpass those of the SOTA models such as U-Net, PraNet, and DS-TransUNet.

TABLE 6. Quantitative results of evaluation metrics for seUNet-Trans in comparison to SOTA models on the 2018 Data Science Bowl.

Methodology	mDC	mIoU	mRec.	mPre.
U-Net [3]	0.757	0.910	-	-
UNet++ [31]	0.897	0.926	-	-
Attention UNet [32]	0.908	0.910	-	0.916
DoubleU-Net [38]	0.913	0.841	0.641	**0.950**
FANet [39]	0.918	0.857	0.922	0.919
DS-TransUNet-L [43]	0.922	0.861	**0.938**	0.912
DCSAU-Net [64]	0.914	0.850	0.924	**0.915**
seUNet-Trans-L (ours)	0.926	0.862	0.894	0.960
seUNet-Trans-M (ours)	**0.928**	**0.867**	0.911	0.947
seUNet-Trans-S (ours)	0.914	0.842	0.884	**0.950**

Specifically, our seUNet-Trans-M attains impressive mDC and mIoU scores of 0.942 and 0.913, respectively, on the Kvasir dataset, and 0.945 and 0.915 on the ClinicDB dataset, as highlighted in Table 7. Remarkably, even on datasets it wasn't explicitly trained on, including ColonDB and EndoScene, our seUNet-Trans demonstrates exceptional predictive accuracy. On ColonDB, it achieves mDC and mIoU of 0.905 and 0.864, respectively, while on EndoScene, these metrics stand at 0.903 and 0.861, showcasing the model's robustness and adaptability.

Sections V-A to V-F conducted a comprehensive comparison of our models with respect SOTA methods across different benchmark medical image datasets. In this section, we aim to present complementary results that intuitively demonstrate the capabilities of our seUNet-Trans models, emphasizing clearer predictions and detection of miss-labeling data.

Figure 11 shows the predictions of our seUNet-Trans on different datasets. Even when dealing with relatively small datasets such as GlaS or the 2018 Data Science Bowl, the model illustrates impressive performance with respect to the ground truth, as shown in Fig. 111. Notably, in the case of the 2018 Data Science Bowl, as depicted in Fig. 11b, our model shows the ability to recognize mislabeling, even when the ground truth does not precisely align with the input image.

Similarly, as illustrated in Fig. 11c, our model also demonstrates its prediction capability on the ISIC2018 dataset. These predictions, generated by the seUNet-Trans-M, closely adhere to the object boundaries within the input images, rather than rigidly following the ground truth. These results indicate the model's proficiency in providing precise predictions across various datasets, thereby promoting its potential as a strong tool for medical image segmentation.

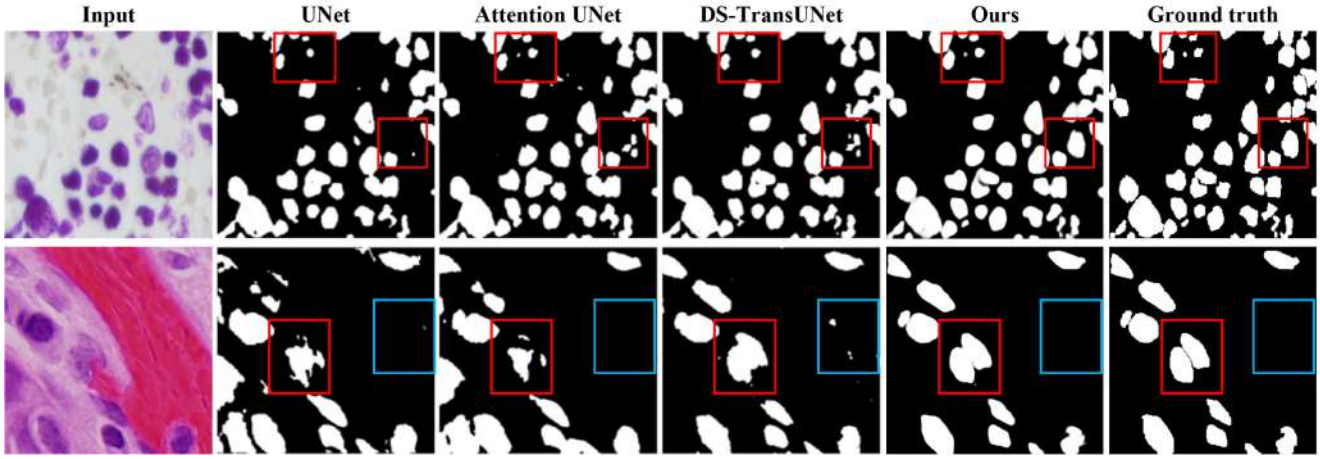


FIGURE 9. Visualization of predictions of seUNet-Trans-M and the SOTA on the 2018 Data Science Bowl dataset. In these images, accurate regions are highlighted with red rectangles, while outliers among the models are indicated with cyan rectangles. These images are partially taken from [43] for comparison purposes.

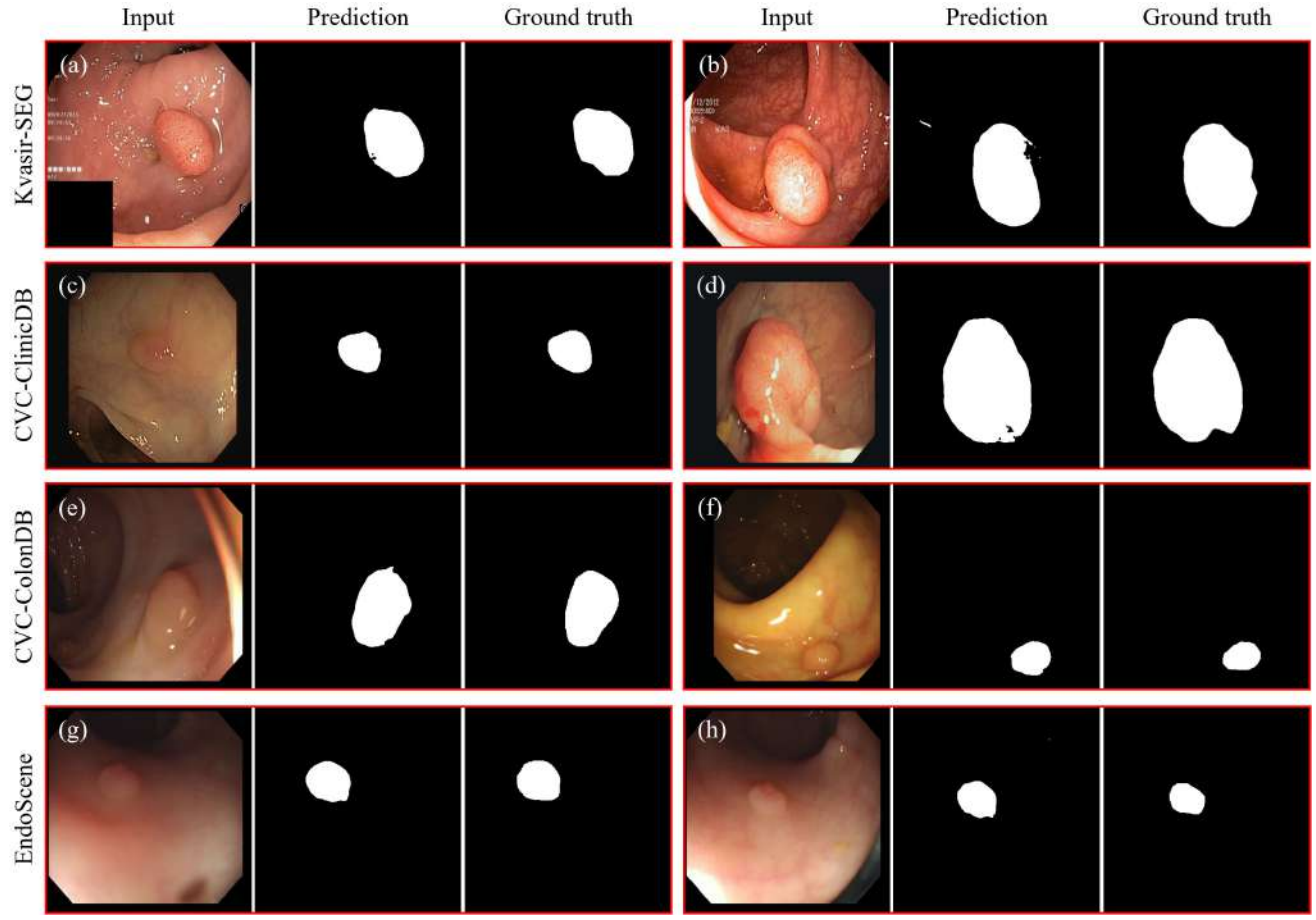
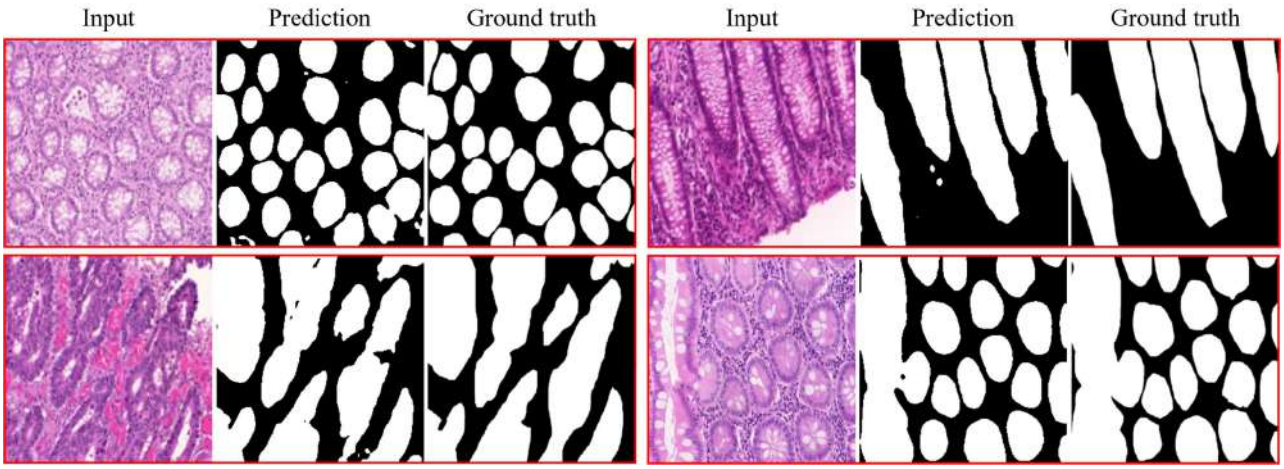


FIGURE 10. Visualization of predictions produced by seUNet-Trans-M in the context of the mixing Polyp Segmentation experiment. Panels (a) and (b) pertain to CVC-ClinicDB, panels (c) and (d) to Kvasir-SEG, panels (e) and (f) to CVC-ColonDB, and panels (g) and (h) to EndoScene representations.

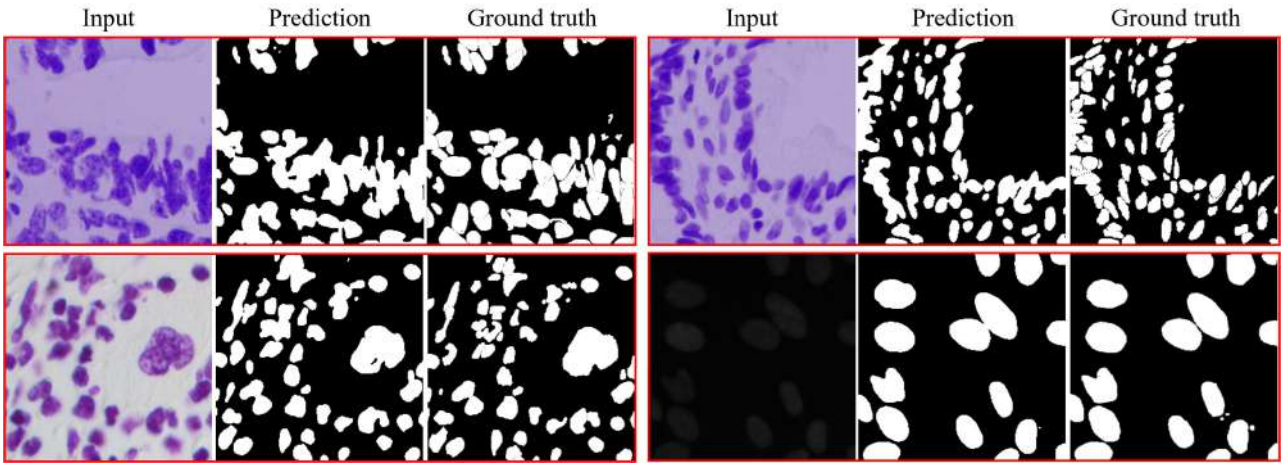
G. ABLATION STUDY

Tables 2-6 present the performance of our models compared to other models. Notably, the seUnet-Trans-M model outperforms both the S and L versions. Consequently, we investigated several bridge layers within the seUnet-Trans-M model. This investigation was conducted across five datasets, as detailed in Table 8.

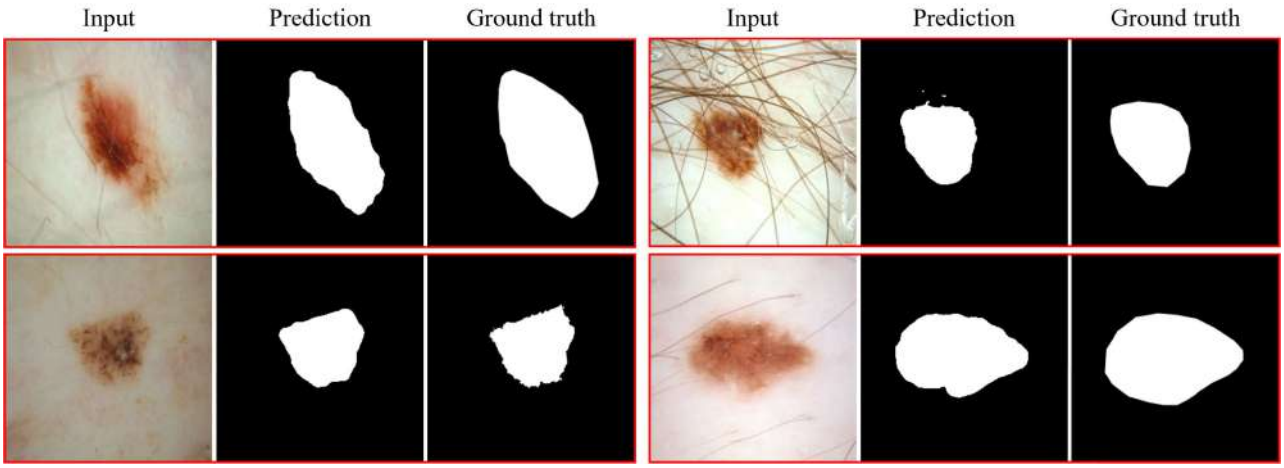
In addition to investigating the importance of the number of bridge layers, we also experimented with low-resolution images to demonstrate the effectiveness of our model.



(a) Visualization of predictions produced by the seUNet-Trans-M on the GlaS dataset.



(b) Visualization of predictions produced by the seUNet-Trans-M on the 2018 Data Science Bowl dataset.



(c) Visualization of predictions produced by the seUNet-Trans-M on the ISIC2018 dataset.

FIGURE 11. Qualitative comparison of different predictions on different datasets by visualization. On a specific set of images covered by a red rectangular, the input, prediction, and ground are organized from the left to the right.

Specifically, we reduced the size of the original images to 128 by 128 and then used them for training. As illustrated in Table 9, the performance of seUNet-Trans-M slightly decreased on the Kvasir-SEG, CVC-ClinicDB, and ISIC 2018 datasets. In the 2018 Data Science Bowl dataset, the performance of our model dropped by around 4.5% compared

TABLE 7. Quantitative results of evaluation metrics for seUNet-Trans in comparison to SOTA models across four different datasets.

Methodology	Kvasir		ClinicDB		ColonDB		EndoScene		Average	
	mDC	mIoU	mDC	mIoU	mDC	mIoU	mDC	mIoU	mDC	mIoU
U-Net	0.818	0.746	0.823	0.755	0.512	0.444	0.398	0.335	0.652	0.581
U-Net++	0.821	0.743	0.794	0.729	0.483	0.410	0.401	0.344	0.641	0.570
PraNet	0.898	0.840	0.899	0.849	0.709	0.640	0.871	0.797	0.800	0.739
HarDNet-MSEG	0.912	0.857	0.932	0.882	0.731	0.660	0.887	0.821	0.828	0.767
TransFuse-L	0.918	0.868	0.934	0.886	0.744	0.676	0.904	0.838	0.847	0.786
DS-TransUNet-L	0.935	0.889	0.936	0.887	0.798	0.722	**0.911**	0.846	0.868	0.806
seUNet-Trans-M (ours)	**0.942**	**0.913**	**0.945**	**0.915**	**0.905**	**0.864**	0.903	**0.861**	**0.934**	**0.899**

TABLE 8. An investigation of the number of bridge layers in the seUNet-Trans model on different datasets.

Dataset	Methods	mDC	mIoU	mRec.	mPre.
Kvasir-SEG	seUNet-Trans-M-16	0.897	0.840	0.887	0.910
	seUNet-Trans-M-32	0.902	**0.852**	0.904	0.896
	seUNet-Trans-M-64	**0.919**	0.850	0.912	**0.926**
	seUNet-Trans-M-128	0.915	0.844	**0.920**	0.922
CVC-ClinicDB	seUNet-Trans-M-16	0.899	0.861	0.850	0.942
	seUNet-Trans-M-32	0.937	0.893	0.933	0.937
	seUNet-Trans-M-64	**0.945**	**0.895**	**0.951**	**0.950**
	seUNet-Trans-M-128	0.934	0.885	0.941	0.940
GlaS	seUNet-Trans-M-16	0.884	0.821	0.884	0.880
	seUNet-Trans-M-32	0.895	0.798	0.881	0.805
	seUNet-Trans-M-64	**0.899**	0.823	**0.886**	**0.920**
	seUNet-Trans-M-128	0.891	**0.830**	0.881	0.899
ISIC 2018	seUNet-Trans-M-16	0.895	**0.854**	0.892	0.924
	seUNet-Trans-M-32	0.913	0.844	**0.920**	0.922
	seUNet-Trans-M-64	**0.922**	**0.854**	0.903	**0.941**
	seUNet-Trans-M-128	0.912	0.847	0.913	0.938
2018 Data Science Bowl	seUNet-Trans-M-16	0.914	0.843	0.884	0.938
	seUNet-Trans-M-32	0.922	0.857	0.932	0.915
	seUNet-Trans-M-64	**0.928**	0.867	0.911	**0.947**
	seUNet-Trans-M-128	0.918	**0.877**	**0.938**	0.922

TABLE 9. Investigation of model's effectiveness (seUNet-Trans-M) with low-resolution images.

Dataset	mDC	mIoU	mRec.	mPre.
Kvasir-SEG	0.907	0.821	0.901	0.918
CVC-ClinicDB	0.930	0.870	0.937	0.942
ISIC 2018	0.918	0.844	0.938	0.909
2018 Data Science Bowl	0.888	0.807	0.895	0.921

to its performance on high-resolution images. These results are due to the cells' nuclei being very small and unclear when using low-resolution images, as shown in Figure 11b. This experiment demonstrates that our model remains robust even with low-resolution images, especially on colonoscopy images.

VI. CONCLUSION AND DISCUSSION

This paper introduces an innovative approach, named seUNet-Trans, that synergizes the robust feature extraction capabilities of CNN with the effective contextual understanding power of Transformer-based models to advance medical image segmentation. Our seUNet-Trans framework employs a hybrid design, integrating a fully convolutional network, UNet, with a Transformer-based model. At the heart of this integration is a specially designed bridge layer that acts as a connection, sequentially channeling rich feature maps from UNet into the Transformer. This design enables the framework to leverage the spatial hierarchies recognized by UNet and the global dependencies discerned by the Transformer, providing a more precise and context-aware segmentation performance.

In our approach, we streamline the architecture of the proposed model by adopting a pixel-level embedding technique that forgoes the traditional use of position embeddings. Furthermore, we explore the trade-offs between computational complexity and model accuracy by employing a computational reduction technique, resulting in the creation of three distinct models (L, M, and S). Such a design enhances the model's efficiency, as it reduces the complexity of the input representation while maintaining the inherent spatial relationships of pixels. This simplification is predicated on

the understanding that, in the context of medical image segmentation, the relative positioning of pixel data is often implicit within the pixel intensity and texture patterns, making separate positional encoding redundant. As a result, our model remains attuned to the crucial spatial cues necessary for accurate segmentation without the computational overhead typically introduced by position embeddings and the original Transformer.

For the investigation of spatially reduced attention, we trained the three models up to 500 epochs, as mentioned in Section IV-B, and compared the training times. Additionally, we compared their inference times on the downstream task. For example, on the CVC-ClinicDB dataset, our seUNet-Trans-L takes around 7 hours for training, while the seUNet-Trans-M and seUNet-Trans-S take approximately 2.46 and 1.53 hours, respectively. During inference, the seUNet-Trans-L takes 4.17 seconds, whereas the seUNet-Trans-M and seUNet-Trans-S take 3.11 and 3.01 seconds, respectively.

We have rigorously evaluated the performance of seUNet-Trans through comprehensive experiments spanning seven diverse datasets. The numerical outcomes from these experiments demonstrate that our proposed model not only meets but also surpasses the benchmarks set by other seminal state-of-the-art models in the majority of the tests. This is particularly notable in mixed Polyp segmentation tasks, where our seUNet-Trans model exhibits superior efficacy, underscoring its robustness and effectiveness in handling complex image segmentation challenges. Although our models achieve good performance, they still struggle with providing separated regions for closely positioned small nuclei.

The promising results presented in this study pave the way for the application of our proposed model across a wider range of tasks. Future endeavors will focus on the development of specialized, lightweight versions of our seUNet-Trans method for specific application needs. Beyond the dataset used, we also plan to investigate the model's ability to handle diverse imaging modalities such as MRI images and 3D brain image segmentation. Moreover, we will investigate the integration of advanced techniques such as the Swin Transformer, which holds the potential to further elevate the efficacy of our model. These explorations are expected to yield significant contributions to the field of medical image analysis and beyond.

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TAN-HANH PHAM received the B.S. degree in mechanical engineering from Ho Chi Minh City University of Technology and Education, Vietnam, in 2019, and the M.S. degree in smart vehicle engineering from Konkuk University, South Korea, in 2022. He is currently pursuing the Ph.D. degree in robotics and deep learning with Florida Institute of Technology, USA. His research interests include deep learning, medical imaging, 3D segmentation, and reinforcement learning in robotics.

XIANQI LI received the M.S. degree in electrical and computer engineering and the Ph.D. degree in applied mathematics from the University of Florida, in 2015 and 2018, respectively. His Ph.D. thesis was on optimization methods for inverse problems.

He is currently working as an Assistant Professor with the Department of Mathematical Sciences, Florida Institute of Technology. Previously, he was a Postdoctoral Research Fellow at the Mass General Hospital, Harvard Medical School, from 2018 to 2021. His research interests include optimization, machine/deep learning, and data analytics, focusing on developing models and algorithms for imaging informatics and data analytics.

KIM-DOANG NGUYEN received the M.S. degree in mechanical engineering from Nanyang Technological University, Singapore, in 2010, and the Ph.D. degree in mechanical engineering from the University of Illinois Urbana-Champaign, in 2015. He was a Postdoctoral Fellow at the University of Technology Sydney, from 2016 to 2017. He is currently working as an Assistant Professor with the Department of Mechanical and Civil Engineering, Florida Institute of Technology. His research interests include robotics, machine learning, controls, and precision agriculture.